Provenance Ledger record certificate

Issued 2026-10-17T20:57:44.227Z by Provenance Ledger

This record shows how this document was written in this environment; it does not detect AI, certify originality, or prove what the author consulted elsewhere.

The record

Record id	`841eae2c738789a517b418031561cc0b4e01901854e83e2e7312fc746e0ab372` (SHA-256 of the record's first event)
Tenant	`tp-06e272def2eb0da0477415b7c1c7e140`
Events	26, in 2 capture session(s), contract ledger-event/v0.1
Chain head	`event 26, 37d41436c24fd0ecff5c5c387a49416774c9da25009ce2795c058bfadb7c58d5`
Head received	2026-10-17T18:13:48.331Z

Where the head is anchored

Sealed hour	2026-10-17T18:00:00.000Z to 2026-10-17T19:00:00.000Z, 51 events of the tenant
Hour root	`4cfc63e451467339d749594dd36af0ec1eae9aa7dcfb8ceac6d07a1374413d81`
Root record SHA-256	`8a271d96a53bab821073eff87d267e6739462f6ee90c3dfaf3b7541b5c86d79a`
Anchored root	`3b0187d9ab5c4eb4e34bbd027ca7fcd58a52238cc6aab4bf91c8b925add77b6a`

Authority	Time
RFC 3161, DigiCert	2026-10-17T19:12:34.000Z
RFC 3161, Sectigo	2026-10-17T19:12:34.000Z
OpenTimestamps, Bitcoin	block 971901, mined 2026-10-17T19:55:21.000Z

Bitcoin block	`971901, hash 00000000000000000006614a392a9b920a3b604e01496aeec809341051ef248`
Merkle root	`fa0a2582916761444d51567138b0c95db332298e829bbb90421329f65b84fa28` (header byte order)

The certificate

Certificate id	`urn:uuid:af6c889d-8204-45e8-bf9f-b4bff6279e1e`
Contract	record-certificate/v0
Issuer key	`Ed25519 d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a`
Issuer key id	`ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e`
Certificate SHA-256	`d43867f906c09cbe4dfb271f47aa7995992606e07fc0f5673aa9858475ec964d` (of certificate.json, exactly as attached)

How to check it

This PDF carries three attached files: certificate.json, the signed record, byte for byte; certificate.sig, the issuer's 64-byte Ed25519 signature of those bytes; and verify.sh, the commands below. Run them in a directory holding the two files, with the authorities' root certificates from anchors/tsa/ of the anchors record for step 5. Each check prints ok or FAIL; nothing in them needs this ledger. The same commands are in certificate.json under procedure. Last, compare the block hash with any Bitcoin node or explorer. The certificate's SHA-256 is listed in the public submission index (submission-index/v0).

```
# Record certificate check: bash with sed, sha256sum, base64, grep, cut, fold, tac and OpenSSL 3.
# In a directory with certificate.json and certificate.sig, and for step 5 the authorities' root
# certificates from anchors/tsa/ of the anchors record (https://github.com/Korentsvit/fundingmachine-anchors).
set -euo pipefail
unhex() { printf "$(tr -d ' \n' | sed 's/../\\x&/g')"; }
step() { printf '%s%s%s' "$1" "$x" "$2" | unhex | sha256sum | cut -c1-64; }
check() { if [ "$1" = "$2" ]; then echo "ok    $3"; else echo "FAIL  $3"; exit 1; fi; }
```

```
field() { grep -o "\"$1\":\"[^\"]*\"" "$2" | sed -n "${3:-1}p" | cut -d'"' -f4; }
echo "certificate SHA-256 $(sha256sum certificate.json | cut -c1-64)"
# 1. The issuer's key signed exactly these bytes.
printf '%s' 302a300506032b6570032100d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a | unhex > issuer.der
openssl pkeyutl -verify -pubin -keyform DER -inkey issuer.der -rawin -in certificate.json -sigfile certificate.sig
check "$(printf '%s' d2f48067c29393ada201fee5de79f502f821e5e3e22e2d84253f043e0ce5a64a | unhex | sha256sum | cut -c1-64)"
ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e '1. the issuer key id'
# 2. The record's head event, 37d41436c24f..., is in the root of the hour 2026-10-17T18:00:00.000Z.
x=37d41436c24fd0ecff5c5c387a49416774c9da25009ce2795c058bfadb7c58d5
x=$(step '00' '')
x=$(step '019e3b76327278975fa1326acc9c10cf57cc4b4b91305bb18715223dcf011289e5' '')
x=$(step '01cbe9e7b1a33282d11b3925eec32cecf335757ff3889d1cc52bed0e7610ce6517' '')
x=$(step '01434c0b1981ae4bfe5152a452894ffe5bf30c141460ee643a87403d31f8b16f3b' '')
x=$(step '01' '552002730d1fef12e3ed6d9abc8f1376e1735d8b678f602eca1e8c50abefe3d5')
x=$(step '0112b81765b0de6579925f47d171a4eb935ed84747cdc166ed6c17524781780 41f' '')
x=$(step '01' 'faff27dfd381a16f1501a4acf82580d4ef1fdc5bfe91851ae4bd1c3dcc9e83c9')
check "$x" 4cfc63e451467339d749594dd36af0ec1eae9aa7dcfb8ceac6d07a1374413d81 '2. the head event is in the hour root'
# 3. The hour's root record carries that root and is signed by the issuer's key.
field root_record_b64 certificate.json | base64 -d > root-record.json
check "$(sha256sum root-record.json | cut -c1-64)" 8a271d96a53bab821073eff87d267e6739462f6ee90c3dfaf3b7541b5c86d79a '3. the hour root
record'
check "$(field root root-record.json)" 4cfc63e451467339d749594dd36af0ec1eae9aa7dcfb8ceac6d07a1374413d81 '3. the record carries the
hour root'
check "$(field key_id root-record.json)" ad0cda7b3d9eb913560f2d8d01ffd5070d9c1d787fe77bdb8654ec7efd13934e '3. the record names the
issuer key'
sed 's/"sig":"[^"]*",//' root-record.json > root-record-unsigned.json
printf 'ledger-root/v0:%s' "$(sha256sum root-record-unsigned.json | cut -c1-64)" > root-message
printf '%s==' "$(field sig root-record.json)" | tr '_-' '/+' | base64 -d > root-record.sig
openssl pkeyutl -verify -pubin -keyform DER -inkey issuer.der -rawin -in root-message -sigfile root-record.sig
# 4. The root record is in the hour's anchored root.
x=8a271d96a53bab821073eff87d267e6739462f6ee90c3dfaf3b7541b5c86d79a
x=$(step '00' '')
check "$x" 3b0187d9ab5c4eb4e34bbd027ca7fcd58a52238cc6aab4bf91c8b925add77b6a '4. the root record is in the anchored root'
# 5. Each RFC 3161 token stamps the anchored root; checked against its authority's root certificate at its time.
field token_b64 certificate.json 1 | base64 -d > digicert.tsr
if [ ! -f digicert.pem ]; then echo 'skip  5. DigiCert: digicert.pem is not here'; elif openssl ts -verify -digest
3b0187d9ab5c4eb4e34bbd027ca7fcd58a52238cc6aab4bf91c8b925add77b6a -in digicert.tsr -CAfile digicert.pem -attime 1792264354 >/dev/null
2>&1; then echo 'ok    5. DigiCert stamped the anchored root at 2026-10-17T19:12:34.000Z'; else echo 'FAIL  5. DigiCert stamped the
anchored root at 2026-10-17T19:12:34.000Z'; exit 1; fi
field token_b64 certificate.json 2 | base64 -d > sectigo.tsr
if [ ! -f sectigo.pem ]; then echo 'skip  5. Sectigo: sectigo.pem is not here'; elif openssl ts -verify -digest
3b0187d9ab5c4eb4e34bbd027ca7fcd58a52238cc6aab4bf91c8b925add77b6a -in sectigo.tsr -CAfile sectigo.pem -attime 1792264354 >/dev/null
2>&1; then echo 'ok    5. Sectigo stamped the anchored root at 2026-10-17T19:12:34.000Z'; else echo 'FAIL  5. Sectigo stamped the
anchored root at 2026-10-17T19:12:34.000Z'; exit 1; fi
# 6. The anchored root reaches the merkle root of Bitcoin block 971901, whose header is below.
x=3b0187d9ab5c4eb4e34bbd027ca7fcd58a52238cc6aab4bf91c8b925add77b6a
x=$(step '' '6c4028badc4275d710d3a4623c31410e')
x=$(step '' 'dd95b27c9a771533')
x=$(step '' '01fbab8094dbb5c49a1d89d40a4f8aee3901d8a6447aee20f7c065d6d1e57469')
x=$(step '6ecc0e4af715bc10b03779e59d78db7628da7f8eaf3e5a856947d432e1df9fc5' '')
x=$(step '' '75707ca0aa3fed017ada14cc77b4964f')
x=$(step '' '65960fa3d573f2f4252e4197418a9f00124a31d6c5b90966d4241e80c6fb1259')
x=$(step '' 'a801a8274bbf42f13722fc13843a56899c744e6fb3c0ba683d69283c3800a168')
x=$(step '3e8a78a3c1a42dc89790f81b63a96e528f4f86565caa7e2412de215b25216b30' '')
x=$(step '6ad3c8a2' 'a6fcfb82b3d69eab')
x=$(step 'a11ce8c08fa914ed15ad5ba9848647674bf47c291275eb8febbfedb66aa48441' '')
x=$(step '' 'ca4bd454c8a9eddc35d0e21df373b676aef71fbf4329e0313c1f5a70d5ea5ae0')
x=$(step '3089c7e787b3ab9f14282200d4292fb059a8f0d5ee6f6aba28a968a81988949d' '')
x=$(step '' 'c6cee155f900dda69f479ce400c7293c1f3c0473824a4145d7754f5af27bf07b')
x=$(step '' '4b779170fc9267c08738c68892852f0b34b850e199db84cbef1337a0eb83c80f')
x=$(step 'da43236c5e1328c1d569199b894463bfe001848a69634b9054aee74d24f05583' '')
x=$(step '' 'ca789ac8d1581e99e76600c1ed315cd08eb36056b0801864f28cdcc96e40f10d')
x=$(step 'b940aa58b6862fb539f043fac3739a15386bba1a8069d016df2a3424b842c778' '')
x=$(step '' '675cf53fdf5742338f3dbff5c88c75ebc84c10f13016af433a9aa343b3085dc6')
x=$(step 'dd8c04d62cd9492a15a49aa5e8edcf58b3054889819380d9ed5db1778b21b750' '')
x=$(step '' '4399d7cab54726f9e8422d0c5854a849d48e0aaed85f16fd518df654bedaa1dd')
x=$(step '00f7becd700291e33e1294969df165c86267e502aecb37c33ebb00e37e6c046f' '')
x=$(step
'0100000001507f12d79897d34ab86b9238b1dc76d19eebe3b4942b31b11fef856279debb9f0000000000feffffff025e160200000000000160014903abd7c5d0ff71c
7166d3a8f83bffabe824a19e0000000000000000226a20' '7bd40e00')
x=$(step '' '')
x=$(step '2dd216e45c7073daf5c6d20d111297037237a807f10bce24720f38723e5db814' '')
x=$(step '' '')
x=$(step '4d59c9ff75c9ea738f2a45ca9b2f57db6b9905fff394dc138acc26cd41945d8b' '')
x=$(step '' '')
x=$(step '' '7342b6dc2cc37e4c4ce190dd8457f3e0540b2030506e99a5128069571f040e33')
x=$(step '' '')
x=$(step '902b438deec3348b1c5cee1a9f0b78559940e293e8726489ef51a5e927680be4' '')
x=$(step '' '')
x=$(step '' 'f25064457fbd6fd4af2f47d1ce42769ef491acf5dc60cb3a612a066fb05a95de')
x=$(step '' '')
```

```
x=$(step '' '54e1ca6f3440d19e5fed8fd5590490efb87ca2dd093155b3d07be3b431456cc3')
x=$(step '' '')
x=$(step '6a98f191ff7cb5325c73be0443cbebd27a0c82375b8a8a1cb67353ba88873e29' '')
x=$(step '' '')
x=$(step '' '44583e849e6e6fd887611a5895bb8f199a1e006d84dc20bb239c36d48ebba0bb')
x=$(step '' '')
x=$(step '' '69ca6b15bed6513d238bbd0403c9467bb3c1b444b3b2fffba195e07e5b2dc2f2')
x=$(step '' '')
x=$(step '' '733721ea7ac34b4e3bcaa0976b4088f7b27591b403aa91bcd04dcd6fea136712')
x=$(step '' '')
x=$(step '' 'b89b6b900273512c5baa38219dc01343ee82ce41c66b4545f23c7649ffb41de1')
x=$(step '' '')
x=$(step '' 'fc351c915909e44bbc4ed30745d068c9429f267e30ab6e32719d4de53db615d1')
x=$(step '' '')
x=$(step '675d4953203de6cb531f4fde29be6bae7f11141e7d0ab9c5020d9856a126efd3' '')
x=$(step '' '')
check "$x" fa0a2582916761444d51567138b0c95db332298e829bbb90421329f65b84fa28 '6. the anchored root reaches the merkle root of block
971901'
h=00204c22c6d13d615d459944a19e36e3ab132643ab4f8d8688ee0100000000000000000000fa0a2582916761444d51567138b0c95db332298e829bbb90421329f65b8
4fa28a9d2d36a93000217c5312e28
check "${h:72:64}" fa0a2582916761444d51567138b0c95db332298e829bbb90421329f65b84fa28 '6. the block header carries that merkle root'
check "$(printf '%s' "$h" | unhex | sha256sum | cut -c1-64 | unhex | sha256sum | cut -c1-64 | fold -w2 | tac | tr -d '\n')"
000000000000000000006614a392a9b920a3b604e01496aeec809341051ef248 '6. the header is block
000000000000000000006614a392a9b920a3b604e01496aeec809341051ef248'
echo "Last, compare block 971901 with any Bitcoin node or explorer: its hash must be
000000000000000000006614a392a9b920a3b604e01496aeec809341051ef248."
```